NIOS Secondary (Class 10) Mathematics (211)

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

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High-Probability Predicted Mock Examination (85 Marks)

Time: 2½ Hours | Maximum Marks: 85

> **Instructions for the Student**:
> Attempt this paper on **Day 6 (4th November)** with a pen, notebook, and geometry box.
> Set a timer for 2.5 hours. Afterwards, check your answers against the complete step-by-step solutions at the end of this document.

SECTION A (Objective Type Questions — 42 Marks)

Q1 to Q17: Multiple Choice Questions (1 Mark Each)

1. The discriminant of the quadratic equation $2x^2 - 4x + 1 = 0$ is:
   (A) 8
   (B) -8
   (C) 16
   (D) 0

2. The 12th term of the AP $3, 7, 11, 15 \ldots$ is:
   (A) 44
   (B) 47
   (C) 48
   (D) 51

3. By selling a watch for ₹660, a shopkeeper makes a profit of 10%. The cost price of the watch was:

(A) ₹600
(B) ₹590
(C) ₹550
(D) ₹726

4. In a circle of radius $5\ \text{cm}$, a tangent PT of length $12\ \text{cm}$ is drawn from external point P. The distance OP is:
   (A) $7\ \text{cm}$
   (B) $13\ \text{cm}$
   (C) $17\ \text{cm}$
   (D) $\sqrt{119}\ \text{cm}$

5. The angle subtended by a diameter at any point on the circumference of a circle is:
   (A) 45°
   (B) 60°
   (C) 90°
   (D) 180°

6. In a cyclic quadrilateral $ABCD$, $\angle B = 85^\circ$. The measure of $\angle D$ is:
   (A) 85°
   (B) 95°
   (C) 105°
   (D) 180°

7. The distance of point $P(-6, 8)$ from the origin is:
   (A) 6
   (B) 8
   (C) 10
   (D) 14

8. The coordinates of the mid-point of line segment joining $(2, -3)$ and $(6, 5)$ are:
   (A) $(4, 1)$
   (B) $(4, 2)$
   (C) $(8, 2)$
   (D) $(3, 1)$

9. The perimeter of a circle of radius $7\ \text{cm}$ is:
   (A) $22\ \text{cm}$
   (B) $44\ \text{cm}$

(C) $154\ \text{cm}$
(D) $88\ \text{cm}$

10. The volume of a cylinder whose radius is $3\ \text{cm}$ and height is $7\ \text{cm}$ is (take $\pi = \frac{22}{7}$):
    (A) $66\ \text{cm}^3$
    (B) $198\ \text{cm}^3$
    (C) $132\ \text{cm}^3$
    (D) $44\ \text{cm}^3$

11. The value of $\tan 45^\circ + \cos 60^\circ$ is:
    (A) 1
    (B) $\frac{3}{2}$
    (C) $\frac{1}{2}$
    (D) 2

12. The value of $\sec^2 45^\circ - \tan^2 45^\circ$ is:
    (A) 0
    (B) 1
    (C) 2
    (D) -1

13. The mean of $4, 6, 8, x, 12$ is 8. The value of x is:
    (A) 8
    (B) 10
    (C) 9
    (D) 7

14. The mode of the data $2, 3, 5, 3, 7, 3, 8, 2$ is:
    (A) 2
    (B) 3
    (C) 5
    (D) 7

15. If the empirical formula connects mode, median, and mean, then Mode equals:
    (A) $3\,\text{Median} - 2\,\text{Mean}$
    (B) $2\,\text{Median} - 3\,\text{Mean}$
    (C) $3\,\text{Mean} - 2\,\text{Median}$
    (D) $\text{Median} - \text{Mean}$

16. A die is thrown once. The probability of getting a number greater than 4 is:
    (A) $\frac{1}{6}$
    (B) $\frac{1}{3}$
    (C) $\frac{1}{2}$
    (D) $\frac{2}{3}$

17. If $P(E) = 0.35$, then $P(\bar{E})$ is:
    (A) 0.65
    (B) 0.75
    (C) 0.35
    (D) 1.35

Q18 to Q27: Objective Questions (2 Marks Each — 1 Mark Per Sub-Part)

1. **Fill in the blanks**:
   (i) The two factors of $x^2 - 7x + 12$ are $(x - 3)$ and `_____`.
   (ii) The value of the polynomial $3x^2 - 5x + 7$ at $x = 1$ is `_____`.

2. **True or False**:
   (i) The roots of $x^2 + 4x + 4 = 0$ are real and equal. (`TRUE / FALSE`)
   (ii) The n-th term of an AP whose first term is a and common difference is d is given by $a + nd$. (`TRUE / FALSE`)

3. **Fill in the blanks**:
   (i) A tangent to a circle intersects it in exactly `_____` point.
   (ii) The lengths of tangents drawn from an external point to a circle are `_____`.

4. **True or False**:
   (i) The angle in a semicircle is an acute angle. (`TRUE / FALSE`)
   (ii) In a cyclic quadrilateral, the sum of opposite angles is $180°$. (`TRUE / FALSE`)

5. **Fill in the blanks**:
   (i) The distance of the point $(0, 5)$ from the x-axis is `_____` units.
   (ii) The midpoint of the line segment joining $(0, 0)$ and $(4, 6)$ is `_____`.

6. **True or False**:
   (i) $\sin^2 60° + \cos^2 60° = 1$. (`TRUE / FALSE`)
   (ii) The value of $\tan 90°$ is 0. (`TRUE / FALSE`)

7. **Fill in the blanks**:
   (i) The total surface area of a cube of edge a is `_____`.
   (ii) The curved surface area of a cylinder of radius r and height h is `_____`.

8. **True or False**:
   (i) The probability of an event can never exceed 1. (`TRUE / FALSE`)
   (ii) Two coins tossed together yield 6 possible outcomes. (`TRUE / FALSE`)

9. **Installment Buying (Fill in the blanks)**:
   A bicycle has a cash price of ₹3,000. Under an installment plan, ₹600 is paid as down payment.
   (i) The unpaid balance is `_____`.
   (ii) If the remaining balance is cleared in 4 equal monthly installments of ₹650, the total interest paid is `_____`.

10. **Match the following**:
Column I	Column II
(i) $\sin 30^\circ$	(a) $\sqrt{3}$
(ii) $\tan 60^\circ$	(b) $\frac{1}{2}$

Q28: Context / Case Study Question (5 Marks — 1 Mark Per Sub-Part)

1. **Case Study**: A mathematics exhibition model displays a circular garden with centre O and radius $10\ \text{m}$. Two watchtowers are set up at points A and B along the boundary. Two straight pathways PA and PB meet at an observation gate P outside the garden. PA and PB are tangents to the circular garden, and $\angle APB = 60^\circ$.
   (i) What is the measure of $\angle OAP$?
   (ii) What is the measure of central angle $\angle AOB$?
   (iii) If $PA = 15\ \text{m}$, what is the length of pathway PB?
   (iv) What type of triangle is ΔPAB?
   (v) What is the perimeter of quadrilateral $OAPB$ if $PA = 15\ \text{m}$ and radius $= 10\ \text{m}$?

SECTION B (Subjective Questions — 43 Marks)

Q29 to Q37: Very Short Answer Questions (2 Marks Each — 18 Marks)

1. Simplify using algebraic identities: $(2x + 5y)^2 - (2x - 5y)^2$.

2. Find the discriminant and hence write the nature of the roots of the equation $3x^2 - 5x + 2 = 0$.
3. Find the 20th term of the AP $4, 9, 14, 19 \ldots$
4. Tangents PA and PB are drawn from point P to a circle with centre O. If $OP = 10\ \text{cm}$ and radius $OA = 6\ \text{cm}$, calculate the length of tangent PA.
5. In the given figure, $ABCD$ is a cyclic quadrilateral. If $\angle A = (2x + 10)^\circ$ and $\angle C = (3x + 20)^\circ$, find the value of x and the measure of $\angle A$.
6. Find the coordinates of the point which divides the line segment joining $A(1, 3)$ and $B(4, 6)$ internally in the ratio $2 : 1$.
7. Find the area of a sector of a circle of radius $6\ \text{cm}$ if the central angle is 60°. (Take $\pi = \frac{22}{7}$).
8. Evaluate: $\frac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$.
9. A card is drawn at random from a pack of 52 playing cards. Find the probability that the card drawn is:
   (i) A king of red colour
   (ii) A spade

Q38 to Q42: Short Answer Questions (3 Marks Each — 15 Marks)

1. Factorise: $x^3 - 8y^3$.
2. Find the ratio in which the point $P(-1, 6)$ divides the line segment joining the points $A(-3, 10)$ and $B(6, -8)$.
3. Construct a circle of radius $3.5\ \text{cm}$. From a point P at a distance of $6.5\ \text{cm}$ from its centre, construct a pair of tangents to the circle.
4. A solid metallic cylinder of radius $4.2\ \text{cm}$ and height $10\ \text{cm}$ is melted and recast into a sphere. Find the radius of the sphere.
5. Calculate the median of the following frequency distribution:

Class Interval	$0 - 10$	$10 - 20$	$20 - 30$	$30 - 40$	$40 - 50$
Frequency (f)	5	8	15	12	10

Q43 to Q44: Long Answer Questions (5 Marks Each — 10 Marks)

1. Solve the following system of linear equations graphically:

$$x + y = 5$$

$$2x - y = 4$$

Find the vertices of the triangle formed by these two lines and the y-axis.

2. A vertical pole of height $60\ \text{m}$ stands on horizontal ground. From the top of the pole, the angles of depression of two cars parked on opposite sides of the pole are observed to be 30° and 45°. Find the distance between the two cars. (Take $\sqrt{3} \approx 1.732$).

COMPLETE STEP-BY-STEP SOLUTIONS & MARKING SCHEME

Section A Solutions

- **Q1**: (A) 8 [$D = (-4)^2 - 4(2)(1) = 16 - 8 = 8$]
- **Q2**: (B) 47 [$a_{12} = 3 + (11)(4) = 47$]
- **Q3**: (A) ₹600 [$CP = \frac{660}{1.1} = 600$]
- **Q4**: (B) $13\ \text{cm}$ [$OP = \sqrt{5^2 + 12^2} = 13$]
- **Q5**: (C) 90°
- **Q6**: (B) 95° [$180 - 85 = 95$]
- **Q7**: (C) 10 [$\sqrt{(-6)^2 + 8^2} = 10$]
- **Q8**: (A) $(4, 1)$ [$(\frac{2+6}{2}, \frac{-3+5}{2}) = (4, 1)$]
- **Q9**: (B) $44\ \text{cm}$ [$2 \times \frac{22}{7} \times 7 = 44$]
- **Q10**: (B) $198\ \text{cm}^3$ [$\frac{22}{7} \times 9 \times 7 = 198$]
- **Q11**: (B) $\frac{3}{2}$ [$1 + \frac{1}{2} = \frac{3}{2}$]
- **Q12**: (B) 1 [Identity $\sec^2\theta - \tan^2\theta = 1$]
- **Q13**: (B) 10 [$\frac{4+6+8+x+12}{5} = 8 \implies 30 + x = 40 \implies x = 10$]
- **Q14**: (B) 3 [Most frequent]
- **Q15**: (A) $3\,\text{Median} - 2\,\text{Mean}$
- **Q16**: (B) $\frac{1}{3}$ [Numbers are $5, 6 \implies \frac{2}{6} = \frac{1}{3}$]
- **Q17**: (A) 0.65 [$1 - 0.35 = 0.65$]
- **Q18**: (i) $(x - 4)$ | (ii) 5 [$3(1) - 5(1) + 7 = 5$]
- **Q19**: (i) TRUE [$D = 16 - 16 = 0$] | (ii) FALSE [Formula is $a + (n-1)d$]
- **Q20**: (i) one | (ii) equal
- **Q21**: (i) FALSE [It is 90°] | (ii) TRUE
- **Q22**: (i) 5 | (ii) $(2, 3)$
- **Q23**: (i) TRUE | (ii) FALSE [$\tan 90^\circ$ is not defined]

- **Q24**: (i) $6a^2$ | (ii) $2\pi rh$
- **Q25**: (i) TRUE | (ii) FALSE [4 outcomes: HH, HT, TH, TT]
- **Q26**: (i) ₹2,400 [$3000 - 600$] | (ii) ₹200 [$4 \times 650 = 2600 \implies 2600 - 2400 = 200$]
- **Q27**: (i) $\rightarrow$ (b) $\frac{1}{2}$ | (ii) $\rightarrow$ (a) $\sqrt{3}$
- **Q28**: (i) 90° | (ii) 120° [$180 - 60$] | (iii) $15\ \text{m}$ [$PA = PB$] | (iv) Equilateral triangle [Since $PA = PB$ and $\angle P = 60^\circ$] | (v) $50\ \text{m}$ [$15 + 15 + 10 + 10 = 50$]

Section B Detailed Solutions

Q29:
Use identity $(a+b)^2 - (a-b)^2 = 4ab$.
Here $a = 2x, b = 5y \implies 4(2x)(5y) = \mathbf{40xy}$.

Q30:
$a = 3, b = -5, c = 2$.
$D = (-5)^2 - 4(3)(2) = 25 - 24 = \mathbf{1}$.
Since $D > 0$, the roots are **real and distinct (unequal)**.

Q31:
$a = 4, d = 5, n = 20$.
$a_{20} = a + (20-1)d = 4 + 19(5) = 4 + 95 = \mathbf{99}$.

Q32:
In right ΔOAP, $\angle OAP = 90^\circ$.
By Pythagoras: $PA = \sqrt{OP^2 - OA^2} = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = \mathbf{8}\ \text{cm}$.

Q33:
Opposite angles sum to 180°:
$(2x + 10) + (3x + 20) = 180 \implies 5x + 30 = 180 \implies 5x = 150 \implies \mathbf{x = 30}$.
$\angle A = 2(30) + 10 = \mathbf{70}^\circ$.

Q34:
$m : n = 2 : 1$, $A(1,3), B(4,6)$.
$x = \frac{2(4)+1(1)}{2+1} = \frac{8+1}{3} = \frac{9}{3} = 3$.
$y = \frac{2(6)+1(3)}{2+1} = \frac{12+3}{3} = \frac{15}{3} = 5$.
Point is $\mathbf{(3, 5)}$.

Q35:
$\text{Area} = \frac{\theta}{360}\pi r^2 = \frac{60}{360} \times \frac{22}{7} \times 6 \times 6 = \frac{1}{6} \times \frac{22}{7} \times 36 = \frac{132}{7} \approx \mathbf{18.86}\ \text{cm}^\mathbf{2}$.

Q36:
Denominator $= \sin^2 30^\circ + \cos^2 30^\circ = 1$.
Numerator $= 5\left(\frac{1}{2}\right)^2 + 4\left(\frac{2}{\sqrt{3}}\right)^2 - (1)^2 = 5\left(\frac{1}{4}\right) + 4\left(\frac{4}{3}\right) - 1 = \frac{5}{4} + \frac{16}{3} - 1 = \frac{15+64-12}{12} = \mathbf{\frac{67}{12}}$.

Q37:
(i) Red Kings $= 2 \implies P = \frac{2}{52} = \mathbf{\frac{1}{26}}$.
(ii) Spades $= 13 \implies P = \frac{13}{52} = \mathbf{\frac{1}{4}}$.

Q38:
$x^3 - (2y)^3 = (x - 2y)(x^2 + 2xy + (2y)^2) = \mathbf{(x - 2y)(x^2 + 2xy + 4y^2)}$.

Q39:
Let ratio be $k : 1$.
Using x-coordinate: $-1 = \frac{k(6)+1(-3)}{k+1} \implies -(k+1) = 6k - 3 \implies -k - 1 = 6k - 3 \implies 7k = 2 \implies k = \frac{2}{7}$.
Ratio is $\mathbf{2 : 7}$.

Q40:
Radius $r = 3.5$ cm, $OP = 6.5$ cm.
Length of tangent $= \sqrt{6.5^2 - 3.5^2} = \sqrt{(6.5 - 3.5)(6.5 + 3.5)} = \sqrt{3 \times 10} = \sqrt{30} \approx \mathbf{5.48}$ cm.

Q41:
Volume of cylinder $=$ Volume of sphere
$\pi r_1^2 h = \frac{4}{3}\pi r_2^3 \implies (4.2)^2 \times 10 = \frac{4}{3} r_2^3$
$17.64 \times 10 = \frac{4}{3} r_2^3 \implies 176.4 \times \frac{3}{4} = r_2^3 \implies 132.3 = r_2^3 \implies r_2 \approx \mathbf{5.1}$ cm.

Q42:
Total $N = 5 + 8 + 15 + 12 + 10 = 50 \implies \frac{N}{2} = 25$.
Cumulative frequencies: $5, 13, 28, 40, 50$.
Median class is $20 - 30$ ($L = 20, cf = 13, f = 15, h = 10$).
$\text{Median} = 20 + \frac{25-13}{15} \times 10 = 20 + \frac{12}{15} \times 10 = 20 + 8 = \mathbf{28}$.

Q43:
- Equation 1 ($x + y = 5$): $(0, 5), (5, 0), (2, 3)$.
- Equation 2 ($2x - y = 4$): $(0, -4), (2, 0), (3, 2)$.
Intersection point: $x = 3, y = 2 \implies \mathbf{(3, 2)}$.
Vertices of triangle formed with y-axis ($x = 0$):
Line 1 hits y-axis at $(0, 5)$; Line 2 hits y-axis at $(0, -4)$; Intersection is $(3, 2)$.
Vertices: $\mathbf{(0, 5)}$, $\mathbf{(0, -4)}$, $\mathbf{(3, 2)}$.

Q44:

Let pole be $AB = 60\ \text{m}$.

- Car 1 at C (elevation 45°): $\tan 45^\circ = \frac{60}{BC} \implies BC = 60\ \text{m}$.
- Car 2 at D (elevation 30°): $\tan 30^\circ = \frac{60}{BD} \implies \frac{1}{\sqrt{3}} = \frac{60}{BD} \implies BD = 60\sqrt{3} = 60 \times 1.732 = 103.92\ \text{m}$.

Total distance between cars:

$CD = BC + BD = 60 + 103.92 = \mathbf{163.92}\ \text{m}$.